

PART IX – TESTING REQUIREMENTS

Students must create a minimum of 10 test cases covering normal input, boundary input, invalid input, empty data, maximum capacity, search success, search failure, update operation, delete operation, and matrix processing. Record all 10 test cases in the accompanying Student Documentation Worksheet.

No.	Feature	Input	Expected Output	Actual Output	Status
1	Add Student	Valid new record	Record added successfully	STUDENT ADDED and student information displayed	PASS
2	Add Student	Duplicate Student ID	System rejects with error message	Student ID already exists.	PASS
3	Display Students	Non-empty array	All active records listed	Existing student records are displayed	PASS
4	Search Student	Existing Student ID	Matching record displayed	Juan Dela Cruz record is displayed	PASS
5	Search Student	Non-existing Student ID	"Record not found" message	Student not found.	PASS
6	Update Student	Valid ID, new data	Record fields updated	STUDENT UPDATED and new information displayed	PASS
7	Delete Student	Valid ID	Record removed, array reindexed	STUDENT DELETED and deletion message displayed	PASS
8	Bounds Checking	Index beyond array capacity	System blocks and reports invalid index	Maximum capacity is 10 students.	PASS
9	Maximum Capacity	Adding when array is full	System reports array full, add rejected	Student capacity is full.	PASS
10	Matrix Processing	Sample resource matrix	Correct row/column totals displayed	D1 = 28, D2 = 25, D3 = 23, D4 = 21, Grand Total = 97	PASS